

# **AIAC LAB 9.1**

**Lab Experiment: Documentation Generation – Automatic documentation and code comments**

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## Problem 1 – Documenting find\_max() in Three Styles

### Question:

Consider the following Python function:

```
def find_max(numbers):  
    return max(numbers)
```

Task: Write documentation for the function in all three formats: (a) Docstring (b) Inline comments (c) Google-style documentation. Critically compare the three approaches. Discuss the advantages, disadvantages, and suitable use cases of each style. Recommend which documentation style is most effective for a mathematical utilities library and justify your answer.

### Answer:

The Python program below documents find\_max() in all three formats and takes the list of numbers from the user.

### #code

```
problem1.py • Python  
  
1  import inspect  
2  
3  
4  # (a) Docstring version  
5  def find_max_docstring(numbers):  
6      """Return the largest number in a list.  
7  
8      Takes a list of numbers and returns the maximum value  
9      using Python's built-in max() function.  
10     """  
11     return max(numbers)  
12  
13  
14  # (b) Inline comments version  
15  def find_max_inline(numbers):  
16      # Use the built-in max() to scan the whole list  
17      # and pick the largest element  
18      return max(numbers) # returns the maximum value  
19  
20
```

```
21 # (c) Google-style documentation version
22 def find_max_google(numbers):
23     """Finds the maximum value in a list of numbers.
24
25     Args:
26         numbers (list): A non-empty list of int or float values.
27
28     Returns:
29         int or float: The largest value in the list.
30
31     Raises:
32         ValueError: If the list is empty.
33
34     Example:
35         >>> find_max_google([3, 7, 2])
36         7
37     """
38     return max(numbers)
39
40
41 # User input
42 values = input("Enter integers separated by spaces: ")
43 numbers = [int(x) for x in values.split()]
44
45 print("Docstring version      :", find_max_docstring(numbers))
46 print("Inline comment version:", find_max_inline(numbers))
47 print("Google style version  :", find_max_google(numbers))
48
49 print("\nDocstring of (a):")
50 print(inspect.getdoc(find_max_docstring))
51 print("\nDocstring of (b):")
52 print(inspect.getdoc(find_max_inline))
53 print("\nDocstring of (c):")
54 print(inspect.getdoc(find_max_google))
```

## #output

```
TERMINAL

PS C:\Users\V\OneDrive\AIAC> python problem1.py
Enter integers separated by spaces: 45 12 89 33 7
Docstring version      : 89
Inline comment version: 89
Google style version   : 89

Docstring of (a):
Return the largest number in a list.

Takes a list of numbers and returns the maximum value
using Python's built-in max() function.

Docstring of (b):
None

Docstring of (c):
Finds the maximum value in a list of numbers.

Args:
  numbers (list): A non-empty list of int or float values.

Returns:
  int or float: The largest value in the list.

Raises:
  ValueError: If the list is empty.

Example:
>>> find_max_google([3, 7, 2])
7
```

## #comparison of the three styles

| Style                      | Advantages                                                                                                                                           | Disadvantages                                                                                                                 | Suitable use cases                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>(a) Docstring</b>       | Stored with the function, so help(), pydoc and IDE tooltips can show it at run time. Short and standard (PEP 257).                                   | A plain summary has no fixed structure, so parameter, return and error details are easy to leave out.                         | Small, simple functions that only need a short description.                                             |
| <b>(b) Inline comments</b> | Sit next to the exact line they explain; good for explaining why something is done.                                                                  | Not available through help() or documentation tools; easily become outdated; a comment on a one-line max() call adds clutter. | Tricky or non-obvious logic inside a function body, mainly for maintainers reading the source.          |
| <b>(c) Google-style</b>    | Clear Args / Returns / Raises / Example sections; easy to read as plain text; Sphinx (napoleon) and IDEs can parse it; examples can run as doctests. | Longer than the code it describes for tiny functions; must be updated whenever behaviour changes.                             | Public APIs, libraries and team projects where other people call the function without reading its code. |

**Recommendation:** Google-style docstrings are the most effective for a mathematical utilities library. Mathematical functions need precise contracts: the expected input types, what happens in edge cases (an empty list raises ValueError), the return type and a worked example. The Args / Returns / Raises / Example sections capture all of this in a fixed structure, the examples double as doctests, and tools such as help(), pydoc, IDE tooltips and Sphinx can turn them into reference pages for the library's users. Inline comments should only supplement them where a line of logic is not obvious.

## Problem 2 – Documenting login()

### Question:

Consider the following Python function:

```
def login(user, password, credentials):  
    return credentials.get(user) == password
```

Task: Write documentation in all three formats. Critically compare the approaches. Recommend which style would be most helpful for new developers onboarding a project, and justify your choice.

### Answer:

The Python program below documents login() in all three formats and takes the username and password from the user.

### #code

```
problem2.py • Python  
  
1  # Demo credentials (username -> password)  
2  credentials = {"admin": "Admin@123", "lakshitha": "Pass@2710", "guest": "guest123"}  
3  
4  
5  # (a) Docstring version  
6  def login_docstring(user, password, credentials):  
7      """Check whether the username and password are valid.  
8  
9      Looks up the user in the credentials dictionary and compares  
10     the stored password with the password entered.  
11     """  
12     return credentials.get(user) == password  
13  
14  
15  # (b) Inline comments version  
16  def login_inline(user, password, credentials):  
17      # get() returns None if the user does not exist, so unknown users fail  
18      # Compare the stored password with the entered one (True = login success)  
19      return credentials.get(user) == password  
20  
21
```

```

problem2.py • Python
22  # (c) Google-style documentation version
23  def login_google(user, password, credentials):
24      """Validates a user's login attempt.
25
26      Args:
27          user (str): Username entered by the user.
28          password (str): Password entered by the user.
29          credentials (dict): Mapping of username (str) to password (str).
30
31      Returns:
32          bool: True if the user exists and the password matches,
33               otherwise False.
34
35      Note:
36          Passwords are compared as plain text (demo only). Real systems
37          should store and compare hashed passwords.
38
39      Example:
40          >>> login_google("admin", "Admin@123", {"admin": "Admin@123"})
41              True
42      """
43      return credentials.get(user) == password
44
45
46  # User input
47  user = input("Enter username: ")
48  password = input("Enter password: ")
49
50  print("Docstring version :", login_docstring(user, password, credentials))
51  print("Inline version    :", login_inline(user, password, credentials))
52  print("Google version    :", login_google(user, password, credentials))
53
54  if login_google(user, password, credentials):
55      print("Login successful")
56  else:
57      print("Login failed")

```

## #output

```

TERMINAL
PS C:\Users\V\OneDrive\AIAC> python problem2.py
Enter username: lakshitha
Enter password: Pass@2710
Docstring version : True
Inline version    : True
Google version    : True
Login successful
PS C:\Users\V\OneDrive\AIAC> python problem2.py
Enter username: admin
Enter password: wrong123
Docstring version : False
Inline version    : False
Google version    : False
Login failed

```

## #comparison of the three styles

| Style         | Advantages                                                              | Disadvantages                                                                                                                    | Suitable use cases                           |
|---------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| (a) Docstring | Explains the purpose of login() in help(login) with very little effort. | A short summary does not say what shape credentials must have or what is returned for an unknown user unless the writer adds it. | Quick description of small helper functions. |

| Style                      | Advantages                                                                                                                                                                                   | Disadvantages                                                                                                         | Suitable use cases                                                        |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>(b) Inline comments</b> | Explain details such as <code>get()</code> returning <code>None</code> for unknown users exactly where the logic sits.                                                                       | A newcomer only sees them after opening the source file; not shown by <code>help()</code> or generated documentation. | Clarifying tricky lines for people already working inside the file.       |
| <b>(c) Google-style</b>    | Args, Returns, Note and Example tell a developer what to pass in (including the credentials dictionary format) and what to expect, without reading the body. Same layout across the project. | Takes longer to write and must be kept in sync with the code.                                                         | Shared or public functions in a team codebase, especially for onboarding. |

**Recommendation:** Google-style docstrings are the most helpful for new developers onboarding a project. A newcomer can call `login()` correctly just by reading its docstring: it shows the type of every argument, the structure of the credentials dictionary, that the result is a `bool` (`False` for an unknown user), a runnable example and a security note about plain-text passwords. The same text appears in `help()`, IDE tooltips and generated HTML documentation, so nobody has to open the source. Inline comments are a good supplement for one specific line, but they are invisible to documentation tools.

### Problem 3 – Calculator (Automatic Documentation Generation)

#### Question:

Task: Design a Python module named `calculator.py` and demonstrate automatic documentation generation.

Instructions: Create a Python module `calculator.py` that includes the following functions, each written with appropriate docstrings:

`add(a, b)` – returns the sum of two numbers

`subtract(a, b)` – returns the difference of two numbers

`multiply(a, b)` – returns the product of two numbers

`divide(a, b)` – returns the quotient of two numbers

Display the module documentation in the terminal using Python's documentation tools. Generate and export the module documentation in HTML format using the `pydoc` utility, and open the generated HTML file in a web browser to verify the output.

#### Answer:

The module `calculator.py` below contains `add()`, `subtract()`, `multiply()` and `divide()`, each with a docstring. A main block takes the numbers and the operation from the user when the file is run directly.

**#code**

```
1  """Calculator module.
2
3  This module provides basic arithmetic operations: addition,
4  subtraction, multiplication and division.
5  """
6
7
8  def add(a, b):
9      """Return the sum of two numbers.
10
11     Args:
12         a: First number.
13         b: Second number.
14
15     Returns:
16         The sum of a and b.
17     """
18     return a + b
19
20
21 def subtract(a, b):
22     """Return the difference of two numbers.
23
24     Args:
25         a: Number to subtract from.
26         b: Number to subtract.
27
28     Returns:
29         The result of a - b.
30     """
31     return a - b
32
33
34 def multiply(a, b):
35     """Return the product of two numbers.
36
37     Args:
38         a: First number.
39         b: Second number.
40
41     Returns:
42         The product of a and b.
43     """
44     return a * b
45
46
```

calculator.py • Python

```
47 def divide(a, b):
48     """Return the quotient of two numbers.
49
50     Args:
51         a: Dividend (number to be divided).
52         b: Divisor (number to divide by).
53
54     Returns:
55         The result of a / b.
56
57     Raises:
58         ZeroDivisionError: If b is zero.
59     """
60     if b == 0:
61         raise ZeroDivisionError("Cannot divide by zero")
62     return a / b
63
64
65 if __name__ == "__main__":
66     a = float(input("Enter first number: "))
67     b = float(input("Enter second number: "))
68     op = input("Enter operation (+, -, *, /): ")
69
70     try:
71         if op == "+":
72             result = add(a, b)
73         elif op == "-":
74             result = subtract(a, b)
75         elif op == "*":
76             result = multiply(a, b)
77         elif op == "/":
78             result = divide(a, b)
79         else:
80             raise ValueError("Invalid operation")
81         print(f"Result: {a:g} {op} {b:g} = {result:g}")
82     except (ValueError, ZeroDivisionError) as error:
83         print("Error:", error)
```

## #output

TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python calculator.py
Enter first number: 15
Enter second number: 3
Enter operation (+, -, *, /): /
Result: 15 / 3 = 5
PS C:\Users\V\OneDrive\AIAC> python calculator.py
Enter first number: 10
Enter second number: 0
Enter operation (+, -, *, /): /
Error: Cannot divide by zero
PS C:\Users\V\OneDrive\AIAC> python calculator.py
Enter first number: 7
Enter second number: 5
Enter operation (+, -, *, /): +
Result: 7 + 5 = 12
```

## #documentation in the terminal (python -m pydoc calculator)

### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python -m pydoc calculator
```

```
Help on module calculator:
```

#### NAME

```
calculator - Calculator module.
```

#### DESCRIPTION

```
This module provides basic arithmetic operations: addition,  
subtraction, multiplication and division.
```

#### FUNCTIONS

```
add(a, b)
```

```
Return the sum of two numbers.
```

```
Args:
```

```
a: First number.
```

```
b: Second number.
```

```
Returns:
```

```
The sum of a and b.
```

```
divide(a, b)
```

```
Return the quotient of two numbers.
```

```
Args:
```

```
a: Dividend (number to be divided).
```

```
b: Divisor (number to divide by).
```

```
Returns:
```

```
The result of a / b.
```

```
Raises:
```

```
ZeroDivisionError: If b is zero.
```

### TERMINAL

```
multiply(a, b)
```

```
Return the product of two numbers.
```

```
Args:
```

```
a: First number.
```

```
b: Second number.
```

```
Returns:
```

```
The product of a and b.
```

```
subtract(a, b)
```

```
Return the difference of two numbers.
```

```
Args:
```

```
a: Number to subtract from.
```

```
b: Number to subtract.
```

```
Returns:
```

```
The result of a - b.
```

#### FILE

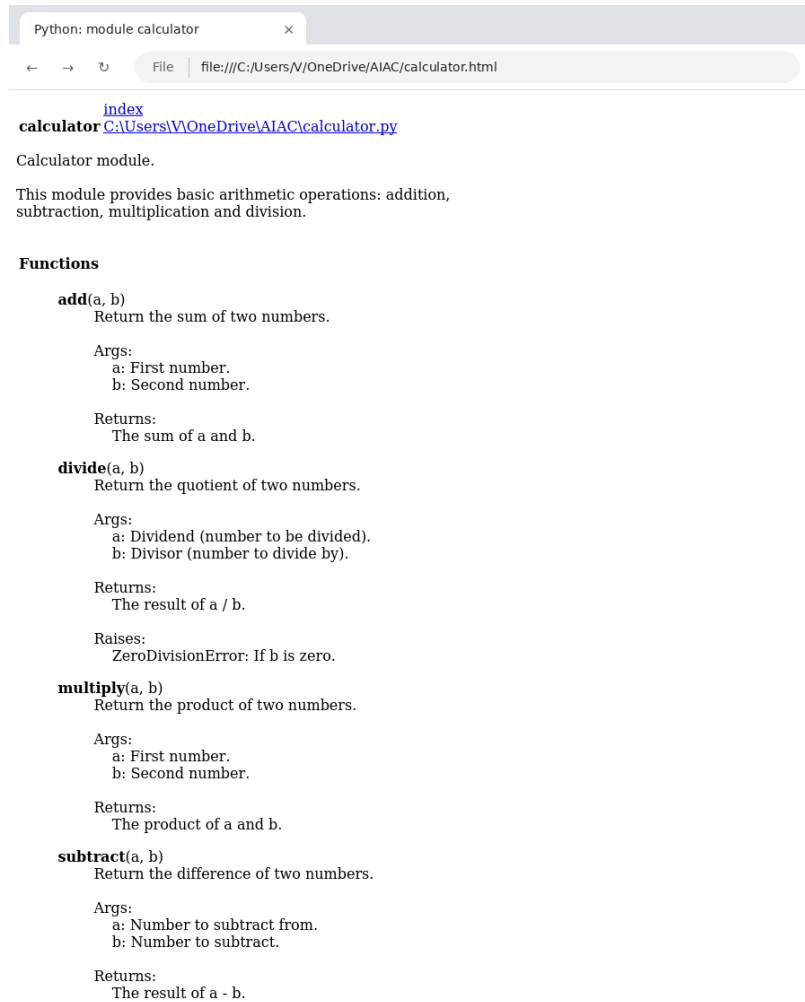
```
C:\Users\V\OneDrive\AIAC\calculator.py
```

## #export documentation as HTML (python -m pydoc -w calculator)

### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python -m pydoc -w calculator
wrote calculator.html
```

## #calculator.html opened in a web browser



The screenshot shows a web browser window with the title "Python: module calculator". The address bar displays the file path: "file:///C:/Users/V/OneDrive/AIAC/calculator.html". The page content includes a link to the "index", the title "calculator" with a link to the source file "C:\Users\V\OneDrive\AIAC\calculator.py", and a description of the "Calculator module". It also lists the basic arithmetic operations provided: addition, subtraction, multiplication, and division. Under the "Functions" section, four functions are detailed: **add(a, b)** (returns the sum), **divide(a, b)** (returns the quotient, raises ZeroDivisionError if b is zero), **multiply(a, b)** (returns the product), and **subtract(a, b)** (returns the difference).

Python: module calculator

← → ↺ File file:///C:/Users/V/OneDrive/AIAC/calculator.html

[index](#)

**calculator** [C:\Users\V\OneDrive\AIAC\calculator.py](#)

Calculator module.

This module provides basic arithmetic operations: addition, subtraction, multiplication and division.

**Functions**

**add(a, b)**  
Return the sum of two numbers.

Args:  
a: First number.  
b: Second number.

Returns:  
The sum of a and b.

**divide(a, b)**  
Return the quotient of two numbers.

Args:  
a: Dividend (number to be divided).  
b: Divisor (number to divide by).

Returns:  
The result of a / b.

Raises:  
ZeroDivisionError: If b is zero.

**multiply(a, b)**  
Return the product of two numbers.

Args:  
a: First number.  
b: Second number.

Returns:  
The product of a and b.

**subtract(a, b)**  
Return the difference of two numbers.

Args:  
a: Number to subtract from.  
b: Number to subtract.

Returns:  
The result of a - b.

## Problem 4 – Conversion Utilities Module

### Question:

Task: Write a module named `conversion.py` with functions: `decimal_to_binary(n)`, `binary_to_decimal(b)`, `decimal_to_hexadecimal(n)`.

Use Copilot for auto-generating docstrings. Generate documentation in the terminal. Export the documentation in HTML format and open it in a browser.

### Answer:

The module `conversion.py` below contains the three conversion functions, each with a docstring. A main block asks the user which conversion to perform and reads the number to convert.

### #code

```
conversion.py • Python
1  """Conversion utilities module.
2
3  This module provides functions to convert numbers between the
4  decimal, binary and hexadecimal number systems.
5  """
6
7
8  def decimal_to_binary(n):
9      """Convert a decimal integer to a binary string.
10
11      Args:
12          n (int): A non-negative decimal integer.
13
14      Returns:
15          str: Binary representation of n (without the '0b' prefix).
16
17      Raises:
18          ValueError: If n is negative.
19
20      Example:
21          >>> decimal_to_binary(10)
22          '1010'
23      """
24      if n < 0:
25          raise ValueError("n must be a non-negative integer")
26      return bin(n)[2:]
27
28
29  def binary_to_decimal(b):
30      """Convert a binary string to a decimal integer.
31
32      Args:
33          b (str): A string made up of only the digits 0 and 1.
34
35      Returns:
36          int: Decimal value of the binary number.
37
38      Raises:
39          ValueError: If b contains characters other than 0 and 1.
40
41      Example:
42          >>> binary_to_decimal("1010")
43          10
44      """
45      if b == "" or any(c not in "01" for c in b):
46          raise ValueError("b must contain only 0 and 1")
47      return int(b, 2)
48
49
```

conversion.py • Python

```
50 def decimal_to_hexadecimal(n):
51     """Convert a decimal integer to a hexadecimal string.
52
53     Args:
54         n (int): A non-negative decimal integer.
55
56     Returns:
57         str: Uppercase hexadecimal representation of n
58             (without the '0x' prefix).
59
60     Raises:
61         ValueError: If n is negative.
62
63     Example:
64         >>> decimal_to_hexadecimal(255)
65         'FF'
66     """
67     if n < 0:
68         raise ValueError("n must be a non-negative integer")
69     return hex(n)[2:].upper()
70
71
72 if __name__ == "__main__":
73     print("1. Decimal to Binary  2. Binary to Decimal  3. Decimal to Hexadecimal")
74     choice = input("Enter your choice: ")
75
76     try:
77         if choice == "1":
78             n = int(input("Enter a decimal number: "))
79             print("Binary:", decimal_to_binary(n))
80         elif choice == "2":
81             b = input("Enter a binary number: ")
82             print("Decimal:", binary_to_decimal(b))
83         elif choice == "3":
84             n = int(input("Enter a decimal number: "))
85             print("Hexadecimal:", decimal_to_hexadecimal(n))
86         else:
87             print("Invalid choice")
88     except ValueError as error:
89         print("Error:", error)
```

## #output

TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python conversion.py
1. Decimal to Binary  2. Binary to Decimal  3. Decimal to Hexadecimal
Enter your choice: 1
Enter a decimal number: 10
Binary: 1010
PS C:\Users\V\OneDrive\AIAC> python conversion.py
1. Decimal to Binary  2. Binary to Decimal  3. Decimal to Hexadecimal
Enter your choice: 2
Enter a binary number: 1010
Decimal: 10
PS C:\Users\V\OneDrive\AIAC> python conversion.py
1. Decimal to Binary  2. Binary to Decimal  3. Decimal to Hexadecimal
Enter your choice: 3
Enter a decimal number: 255
Hexadecimal: FF
PS C:\Users\V\OneDrive\AIAC> python conversion.py
1. Decimal to Binary  2. Binary to Decimal  3. Decimal to Hexadecimal
Enter your choice: 2
Enter a binary number: 1021
Error: b must contain only 0 and 1
```

## #documentation in the terminal (python -m pydoc conversion)

### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python -m pydoc conversion
Help on module conversion:
```

#### NAME

conversion - Conversion utilities module.

#### DESCRIPTION

This module provides functions to convert numbers between the decimal, binary and hexadecimal number systems.

#### FUNCTIONS

binary\_to\_decimal(b)

Convert a binary string to a decimal integer.

Args:

b (str): A string made up of only the digits 0 and 1.

Returns:

int: Decimal value of the binary number.

Raises:

ValueError: If b contains characters other than 0 and 1.

Example:

```
>>> binary_to_decimal("1010")
10
```

### TERMINAL

decimal\_to\_binary(n)

Convert a decimal integer to a binary string.

Args:

n (int): A non-negative decimal integer.

Returns:

str: Binary representation of n (without the '0b' prefix).

Raises:

ValueError: If n is negative.

Example:

```
>>> decimal_to_binary(10)
'1010'
```

decimal\_to\_hexadecimal(n)

Convert a decimal integer to a hexadecimal string.

Args:

n (int): A non-negative decimal integer.

Returns:

str: Uppercase hexadecimal representation of n (without the '0x' prefix).

Raises:

ValueError: If n is negative.

Example:

```
>>> decimal_to_hexadecimal(255)
'FF'
```

#### FILE

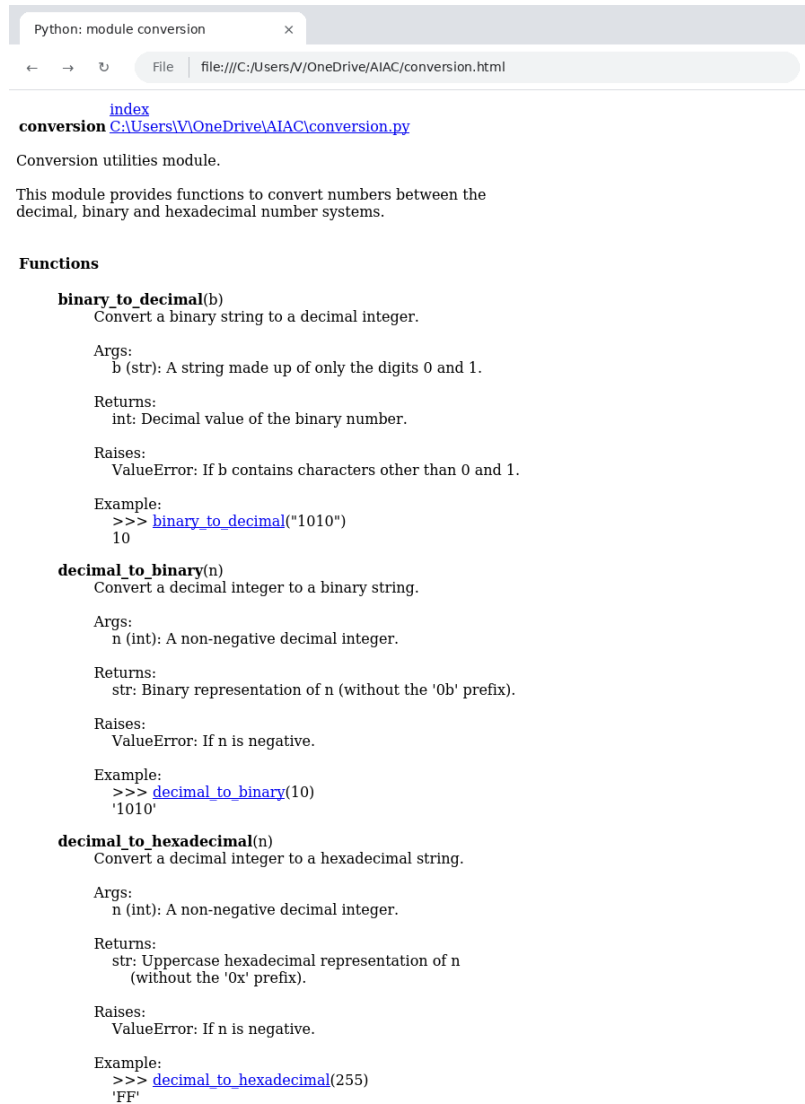
C:\Users\V\OneDrive\AIAC\conversion.py

## #export documentation as HTML (python -m pydoc -w conversion)

### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python -m pydoc -w conversion
wrote conversion.html
```

## #conversion.html opened in a web browser



The screenshot shows a web browser window with a single tab titled "Python: module conversion". The address bar displays the file path: "file:///C:/Users/V/OneDrive/AIAC/conversion.html". The page content includes a navigation link for "index", a title "conversion C:\Users\V\OneDrive\AIAC\conversion.py", and a description: "Conversion utilities module." It also states: "This module provides functions to convert numbers between the decimal, binary and hexadecimal number systems." Under the "Functions" section, three functions are detailed:   
1. **binary\_to\_decimal(b)**: Convert a binary string to a decimal integer. It lists the argument 'b' as a string of 0s and 1s, the return as an integer, and the exception 'ValueError' for invalid characters. An example shows the conversion of "1010" to 10.   
2. **decimal\_to\_binary(n)**: Convert a decimal integer to a binary string. It lists the argument 'n' as a non-negative integer, the return as a string without the '0b' prefix, and the exception 'ValueError' for negative values. An example shows the conversion of 10 to '1010'.   
3. **decimal\_to\_hexadecimal(n)**: Convert a decimal integer to a hexadecimal string. It lists the argument 'n' as a non-negative integer, the return as an uppercase string without the '0x' prefix, and the exception 'ValueError' for negative values. An example shows the conversion of 255 to 'FF'.

Python: module conversion

file:///C:/Users/V/OneDrive/AIAC/conversion.html

[index](#)

**conversion** C:\Users\V\OneDrive\AIAC\conversion.py

Conversion utilities module.

This module provides functions to convert numbers between the decimal, binary and hexadecimal number systems.

**Functions**

**binary\_to\_decimal(b)**  
Convert a binary string to a decimal integer.

Args:  
b (str): A string made up of only the digits 0 and 1.

Returns:  
int: Decimal value of the binary number.

Raises:  
ValueError: If b contains characters other than 0 and 1.

Example:  
>>> [binary\\_to\\_decimal](#)("1010")  
10

**decimal\_to\_binary(n)**  
Convert a decimal integer to a binary string.

Args:  
n (int): A non-negative decimal integer.

Returns:  
str: Binary representation of n (without the '0b' prefix).

Raises:  
ValueError: If n is negative.

Example:  
>>> [decimal\\_to\\_binary](#)(10)  
'1010'

**decimal\_to\_hexadecimal(n)**  
Convert a decimal integer to a hexadecimal string.

Args:  
n (int): A non-negative decimal integer.

Returns:  
str: Uppercase hexadecimal representation of n (without the '0x' prefix).

Raises:  
ValueError: If n is negative.

Example:  
>>> [decimal\\_to\\_hexadecimal](#)(255)  
'FF'

## Problem 5 – Course Management Module

### Question:

Task: Create a module `course.py` with functions: `add_course(course_id, name, credits)`, `remove_course(course_id)`, `get_course(course_id)`.

Add docstrings with Copilot. Generate documentation in the terminal. Export the documentation in HTML format and open it in a browser.

### Answer:

The module `course.py` below stores courses in memory and provides `add_course()`, `remove_course()` and `get_course()`, each with a docstring. A menu-driven main block takes the course details from the user.

### #code

```
course.py • Python

1  """Course management module.
2
3  This module stores courses in memory and provides functions to
4  add, remove and look up courses by their course ID.
5  """
6
7  courses = {}
8
9
10 def add_course(course_id, name, credits):
11     """Add a new course to the course list.
12
13     Args:
14         course_id (str): Unique ID of the course, e.g. "CS101".
15         name (str): Name of the course.
16         credits (int): Number of credits for the course.
17
18     Returns:
19         bool: True if the course was added, False if the
20             course_id already exists.
21     """
22     if course_id in courses:
23         return False
24     courses[course_id] = {"name": name, "credits": credits}
25     return True
26
27
28 def remove_course(course_id):
29     """Remove a course from the course list.
30
31     Args:
32         course_id (str): ID of the course to remove.
33
34     Returns:
35         bool: True if the course was removed, False if it
36             does not exist.
37     """
38     if course_id not in courses:
39         return False
40     del courses[course_id]
41     return True
42
43
```

```
44 def get_course(course_id):
45     """Get the details of a course.
46
47     Args:
48         course_id (str): ID of the course to look up.
49
50     Returns:
51         dict or None: A dictionary with the keys "name" and
52         "credits", or None if the course does not exist.
53     """
54     return courses.get(course_id)
55
56
57 if __name__ == "__main__":
58     while True:
59         print("\n1. Add course 2. Remove course 3. Get course 4. Exit")
60         choice = input("Enter your choice: ")
61
62         if choice == "1":
63             course_id = input("Enter course ID: ")
64             name = input("Enter course name: ")
65             credits = int(input("Enter credits: "))
66             if add_course(course_id, name, credits):
67                 print("Course added")
68             else:
69                 print("Course ID already exists")
70         elif choice == "2":
71             course_id = input("Enter course ID: ")
72             if remove_course(course_id):
73                 print("Course removed")
74             else:
75                 print("Course not found")
76         elif choice == "3":
77             course_id = input("Enter course ID: ")
78             course = get_course(course_id)
79             if course:
80                 print("Name:", course["name"], "| Credits:", course["credits"])
81             else:
82                 print("Course not found")
83         elif choice == "4":
84             print("Exiting course manager")
85             break
86         else:
87             print("Invalid choice")
```

## #output

### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python course.py
```

```
1. Add course  2. Remove course  3. Get course  4. Exit
```

```
Enter your choice: 1
```

```
Enter course ID: CS101
```

```
Enter course name: Python Programming
```

```
Enter credits: 4
```

```
Course added
```

```
1. Add course  2. Remove course  3. Get course  4. Exit
```

```
Enter your choice: 1
```

```
Enter course ID: MA201
```

```
Enter course name: Discrete Mathematics
```

```
Enter credits: 3
```

```
Course added
```

```
1. Add course  2. Remove course  3. Get course  4. Exit
```

```
Enter your choice: 3
```

```
Enter course ID: CS101
```

```
Name: Python Programming | Credits: 4
```

```
1. Add course  2. Remove course  3. Get course  4. Exit
```

```
Enter your choice: 2
```

```
Enter course ID: MA201
```

```
Course removed
```

```
1. Add course  2. Remove course  3. Get course  4. Exit
```

```
Enter your choice: 3
```

```
Enter course ID: MA201
```

```
Course not found
```

```
1. Add course  2. Remove course  3. Get course  4. Exit
```

```
Enter your choice: 4
```

```
Exiting course manager
```

## #documentation in the terminal (python -m pydoc course)

### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python -m pydoc course
```

```
Help on module course:
```

#### NAME

```
course - Course management module.
```

#### DESCRIPTION

```
This module stores courses in memory and provides functions to  
add, remove and look up courses by their course ID.
```

#### FUNCTIONS

```
add_course(course_id, name, credits)
```

```
    Add a new course to the course list.
```

##### Args:

```
    course_id (str): Unique ID of the course, e.g. "CS101".
```

```
    name (str): Name of the course.
```

```
    credits (int): Number of credits for the course.
```

##### Returns:

```
    bool: True if the course was added, False if the  
          course_id already exists.
```

#### TERMINAL

```
get_course(course_id)
    Get the details of a course.

    Args:
        course_id (str): ID of the course to look up.

    Returns:
        dict or None: A dictionary with the keys "name" and
            "credits", or None if the course does not exist.

remove_course(course_id)
    Remove a course from the course list.

    Args:
        course_id (str): ID of the course to remove.

    Returns:
        bool: True if the course was removed, False if it
            does not exist.
```

#### DATA

```
courses = {}
```

#### FILE

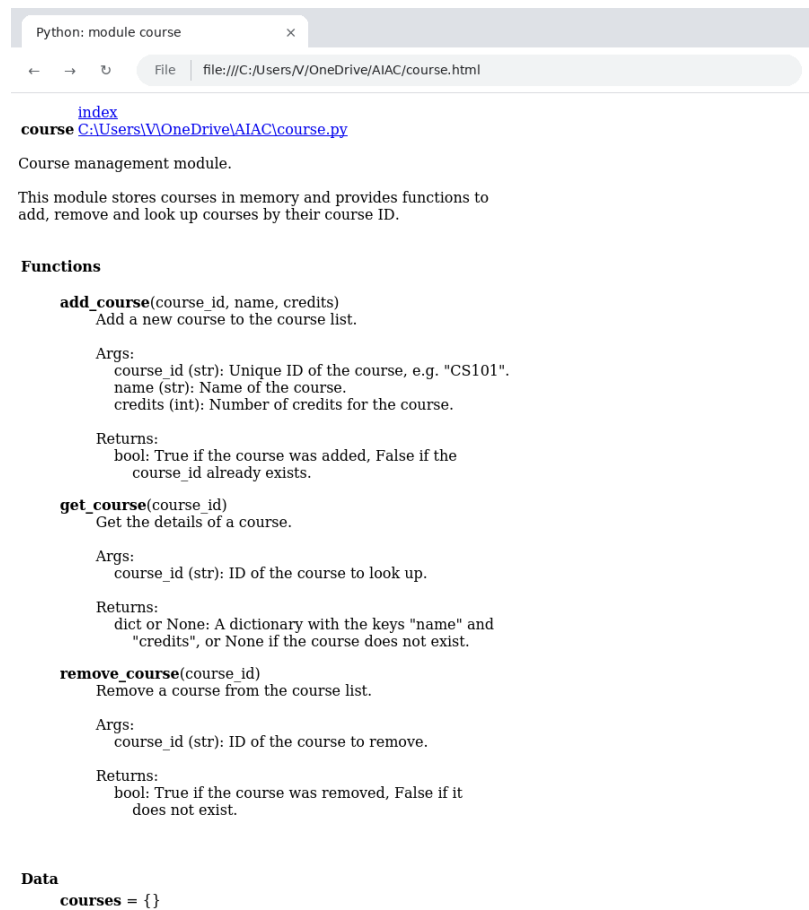
```
C:\Users\V\OneDrive\AIAC\course.py
```

**#export documentation as HTML (python -m pydoc -w course)**

#### TERMINAL

```
PS C:\Users\V\OneDrive\AIAC> python -m pydoc -w course
wrote course.html
```

## #course.html opened in a web browser



## Conclusion

All five problems were completed. `find_max()` and `login()` were documented using docstrings, inline comments and Google-style documentation and the styles were compared. The modules `calculator.py`, `conversion.py` and `course.py` were documented with docstrings, viewed in the terminal with `pydoc`, exported as HTML pages with `pydoc -w` and verified in a web browser, as required in Lab 9.1.